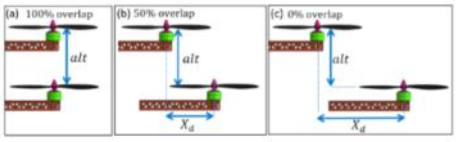
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Wednesday, August 5, 2026 8:36 AM

https://dl.acm.org/doi/epdf/10.1145/3636534.3649382: AeroBridge: Autonomous Drone Handoff System for Emergency Battery Service

- An emergency battery service drone files to a drone with a depleted battery and transfers a fresh battery to the exhausted drone
- Mitigates the effects of aerodynamic downwash between the two drones through diagonal alignment
    - Able to get the distance down to 0.5 m
    - CFD analysis shows that diagonal vs perpendicular alignment minimizes turbulence
        - **Turbulence of 4 m/s found and does not affect the transfer process**
- Can handle three batteries in a single flight
- **Challenges**
    - Downwash caused by drone propellers in vertical flight
        - Heavily affects the stability of the two drones making vertical transfer mechanisms difficult
        - States that post drones have vertical aligned and hover at a distance of 4.5 meters with basic PID calibration and thrust adjustment
        - Finds that downwash is driven by two mechanisms
            - Drag resulting from oncoming flow on the frame
            - Alteration in propeller thrust because of oncoming flow
            - 

            - Figure 2: The drone position model based on air flow position (a) $Xd = 0$ cm (b) $Xd = 16$ cm (c) $Xd = 32$ cm.
    - Positioning the drones with stability and accuracy
        - Requires precise pose and location
        - Subject to effects of wind and imperfect motor actuation of propellers
    - Transfer quickly and minimal disruption of the drones tasks
- **Solutions to challenges**
    - Downwash
        - 0.5 m drone proximity by using diagonal alignment
        - Found ideal optimal outcome as Xd = 16 cm and alt = 60 cm
    - Positioning
        - Vision based algorithm to localize, position, and transfer with downward camera
        - Initial position by GPS and close proximity positioning using a marker based visual odometry technique
        - CMP model for visual inertial navigation
            - Cross marker position model optimizes by correct orientation angles using marker-based pose estimate
        - Experimentally tested exact distances, positions, and angles between the drones
    - Quick transfer
        - Aerobridge design
- **Airflow estimation calculations - 13 inch prop at 5000 rpm**
    - **Downwash at 14m/s at 15 cm with 100% overlap**

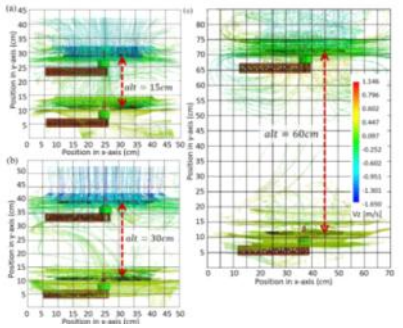
heat map. Based on the blade element theory [4], thrust can be calculated using mass flow rate ($\dot{m}$), thrust (T) for downstream (T_d) and upstream (T_u), and air velocity distribution (v) similarly for downstream (v_d) and upstream (v_u), as described by equations given below:

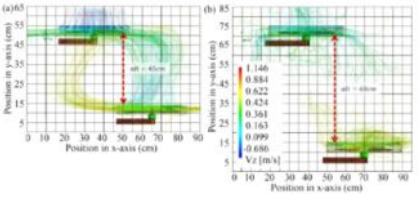
$$T = T_d - T_u = \dot{m}(v_d - v_u) = \dot{m}[(v_d) - (v_\infty)] \quad (1)$$

Now, in equation below (A) is surface area of propeller, (v_i) is induced velocity and (v_∞) is free stream velocity, and air density (ρ).

$$\dot{m} = \rho A[(v_\infty) + (v_i)] \quad (2)$$

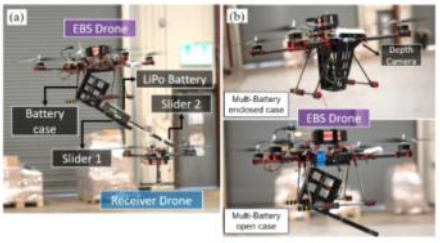
The subsequent section shows the effects of the downwash from the flow rate to the drone below, highlighting unstable airflow turbulence.

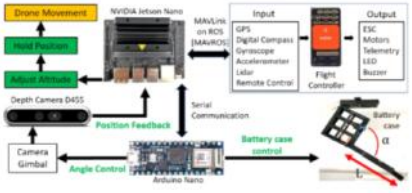


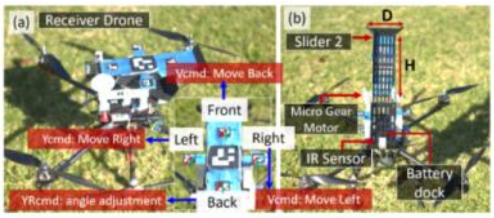


- Verified findings through physical system testing
  - Finalized a vertical separation distance of 0.5 m and 45 degrees angle which is roughly a 50% overlap of the drone dimensions

- **Drone architecture**
  - Quadcopter with Nvidia Jetson Nano
    - Nano is responsible for controlling flight modes and trajectory of the drone
  - 695 x 695 x 210 mm with 13-inch props and 350 KV motors
  - Maximum lifting weight of 4.8 kg
  - Gravity assisted slider mechanism
    - The servo turns the slider to 45 degrees and the batters are free sliding through guided railings



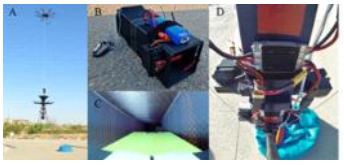


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- Airbridge mechanism

https://ieeexplore.ieee.org/abstract/document/9826886/citations?tabFilter=papers#citations

- For battery replacement swapping consisting of two drones
- Both drones are octocopters
- Delivery drone
  - A cable to lower the battery to the receiver drone
  - Does not go into detail about electronic components
- Receiver drone
  - Tarot X8 octocopter
  - 1.45 m diameter, 1.15 m height, 7.94 kg weight
  - **With packages, max weight was 10.4**
  - **Tested 15 in and 17 in props**
    - Used 15 in for more agility
  - 250 W powered motors
- Main system architecture principles
  - Funnel allows for less alignment precision needed
  - **Sender drone lowers the batteries via cable at 4.5 m**
- 

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https://www.nature.com/articles/s41586-025-09575-x

- FlyingToolbox drone: vertical-stack flying drone that autonomously docks tools into a toolbox drone

Challenges

- High accuracy docking
- Downwash disturbance
  - Found that when the two drones have a distance of **0.6 m, the downwash speed can reach up to 13.18 m/s, exerting 24.9 N (5.6 lbf or 2.54 kgf)**
- Global positioning

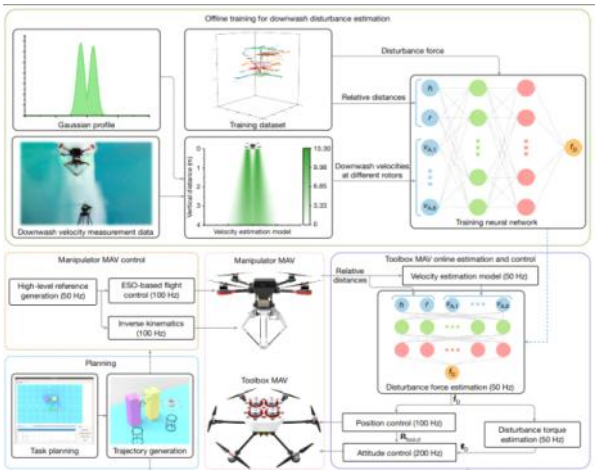
Solutions

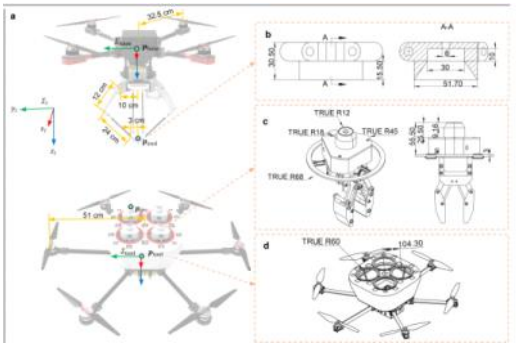
- High accuracy docking
  - Electromagnetic tips with soft robotic ends (much more precise than we should need)
- Downwash disturbance
  - They model the downwash velocity field as a Gaussian profile which allows them to predict the airflow velocity at arbitrary positions below the sender drone
  - Uses a neural network model with relative distance and downwash velocity as input information
  - Able to remain in the downwash but predict its loading to adjust accordingly
- Global positioning
  - External vision positional system was used
  - AprilTag array on the receiver drone and visual positioning system on the sender drone

System architecture

- Senser drone
  - Quadcopter with 3 DOF delta robotic arm
  - Electro magnet at the end of arm to link the tool and arm together with attraction force of 80 N
  - Wireless controller and battery on the tool to allow electromagnet to be wirelessly activated or deactivated
- Receiver drone
  - Hexacopter with slots for the tools







Summary:

- System design challenges
  - Downwash interaction
    - Possible mitigations:
      - Off center alignment
      - Large separation distance
      - High stability drone
  - Positioning and orientation
    - Things to address:
      - horizontal, vertical, depth alignment
      - Required yaw, pitch, roll alignment
    - Different mechanisms
      - GPS
      - Visual detection
      - Relative localization
      - Mechanical alignment
  - Transfer mechanism
    - Things to consider
      - How active is the transfer?
        - Gravity-assisted? Robotically constrained? Mechanically guided?
      - Are the drones attaching themselves?
      - Minimize weight/deflection on drone
      - Minimize moments generated
      - Stability effects of payload weight transfer
  - Wind factor
    - Mitigate effects
      - Cantilever beam deflection
        - Stiff
        - Short
        - Moment of inertia
- Hardware challenges
  - Rotor layout
  - Propulsion and electrical power
  - Communication
  - Payload detection and transfer verification